Area of Triangles

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

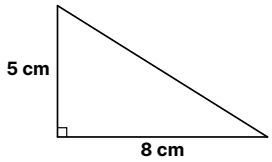
KEY WORDS area *(horahanga)* · triangle *(tapatoru)* · base *(pūtake)* · perpendicular height *(teitei hāngai)*

A Base and height ●○○

Remember: Area = $\frac{1}{2}$ × base × height. The height is measured at right angles to the base, shown by the small square.

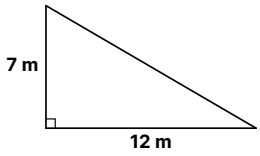
1. Find the area of each triangle.

a)



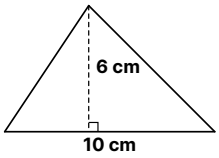
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b)



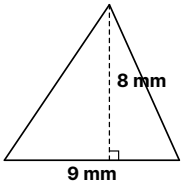
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c)



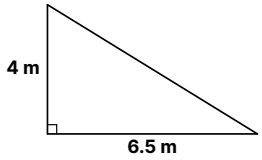
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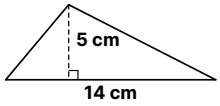
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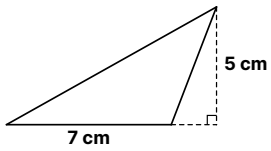
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B Obtuse triangles and extra lengths ●●○

Watch out: Use the perpendicular height, even when it stands outside the triangle. Ignore the slanting sides: they are not the height.

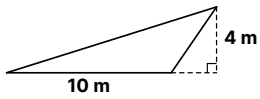
2. Find the area of each triangle.

a)



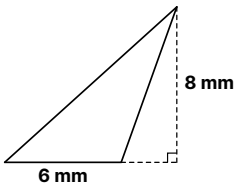
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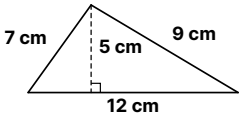
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c)



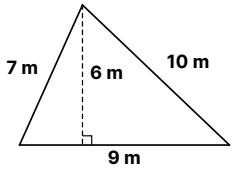
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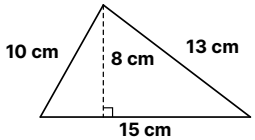
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C Finding a missing length ●●○

Remember: Put what you know into Area = $\frac{1}{2}$ × base × height, then solve for the missing length.

3. Find the missing length.

a) A triangle has an area of 24 cm² and a base of 8 cm. Find its height.

b) A triangle has an area of 35 m² and a height of 7 m. Find its base.

c) A triangle has an area of 45 cm² and a base of 15 cm. Find its height.

d) A triangle has an area of 27 mm² and a height of 9 mm. Find its base.

e) A triangle has an area of 30 m² and a base of 5 m. Find its height.

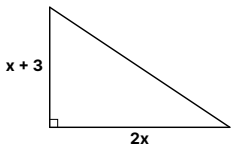
f) A triangle has an area of 52 cm² and a height of 8 cm. Find its base.

D Lengths with algebra ●●●

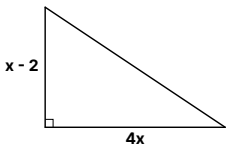
Remember: Treat the expressions like numbers: write $\frac{1}{2}$ × base × height, then expand and simplify.

4. Write an expression for the area of each triangle. Expand and simplify it.

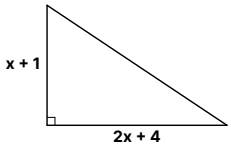
a)



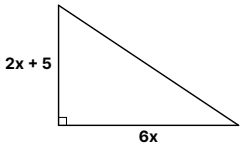
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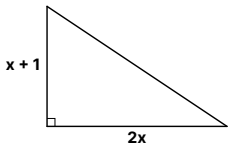


d)



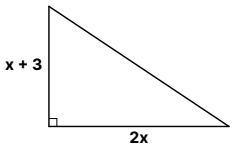
5. The area of each triangle is given. Write an equation and solve it to find x.

a)



The area is 30 cm². Find x.

b)



The area is 28 m². Find x.

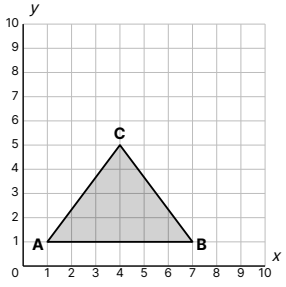
E Problems ●●●

6. A triangular sail has a base of 4.5 m and a height of 6 m. Sailcloth costs $32 per square metre. Find the cost of the cloth for the sail.

7. A rectangle 12 cm by 8 cm has a triangle cut from it. The triangle has its base along the whole bottom side of the rectangle and its top corner 5 cm up the left side. Find the area that is left.

8. A triangle has corners at A(1, 1), B(7, 1) and C(4, 5).

a) Find its area in square units.



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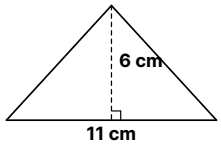
9. A triangular garden bed has a base of 15 m and a height of 8 m. One bag of mulch covers 4 m². How many bags are needed?

F Mixed practice ●●○

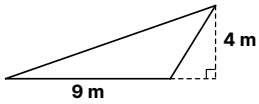
> **Remember:** Every kind of triangle from this sheet, in one go. Pick out the base and the perpendicular height first.

10. Find the area of each triangle.

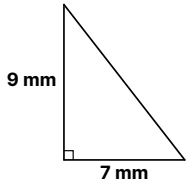
a)



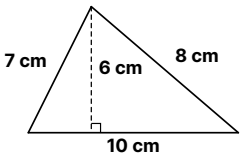
b)



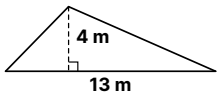
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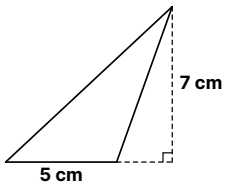
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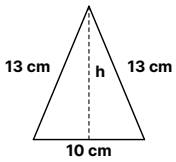


f)



★ Extension

11. An isosceles triangle has two sides of 13 cm and a base of 10 cm. Find its height first, then its area.



12. An equilateral triangle has sides of 10 cm. Find its area, to 1 decimal place.

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Area of Triangles

Name _____ Date _____ Calculator allowed Version 2

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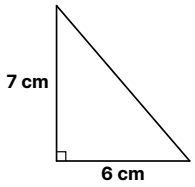
KEY WORDS area *(horahanga)* · triangle *(tapatoru)* · base *(pūtake)* · perpendicular height *(teitei hāngai)*

A Base and height ●○○

Remember: Area = $\frac{1}{2}$ × base × height. The height is measured at right angles to the base, shown by the small square.

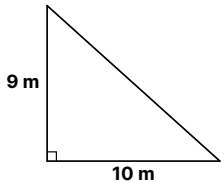
1. Find the area of each triangle.

a)



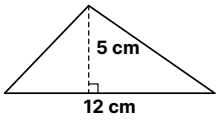
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b)



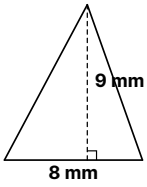
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c)



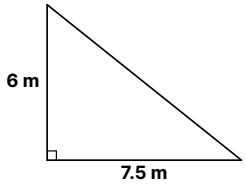
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d)



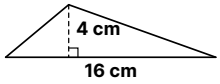
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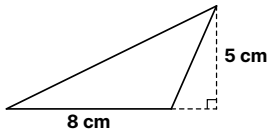
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B Obtuse triangles and extra lengths ●●○

Watch out: Use the perpendicular height, even when it stands outside the triangle. Ignore the slanting sides: they are not the height.

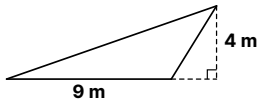
2. Find the area of each triangle.

a)



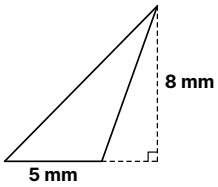
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b)



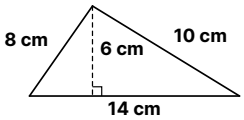
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c)



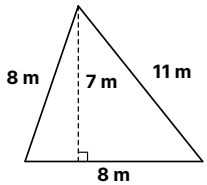
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d)



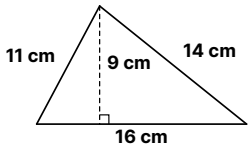
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e)



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f)



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C Finding a missing length ●●○

Remember: Put what you know into Area = $\frac{1}{2}$ × base × height, then solve for the missing length.

3. Find the missing length.

a) A triangle has an area of 28 cm² and a base of 8 cm. Find its height.

b) A triangle has an area of 40 m² and a height of 8 m. Find its base.

c) A triangle has an area of 42 cm² and a base of 14 cm. Find its height.

d) A triangle has an area of 33 mm² and a height of 11 mm. Find its base.

e) A triangle has an area of 36 m² and a base of 6 m. Find its height.

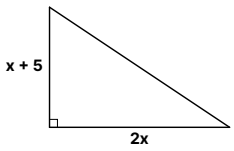
f) A triangle has an area of 63 cm² and a height of 9 cm. Find its base.

D Lengths with algebra ●●●

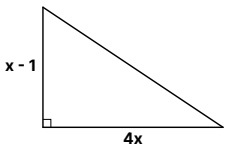
Remember: Treat the expressions like numbers: write $\frac{1}{2}$ × base × height, then expand and simplify.

4. Write an expression for the area of each triangle. Expand and simplify it.

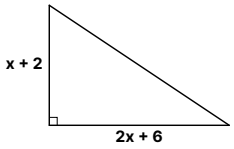
a)



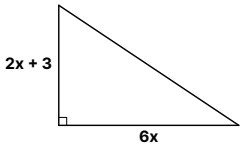
b)



c)

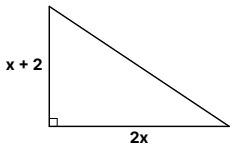


d)



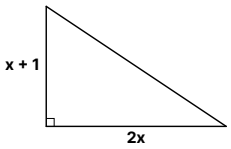
5. The area of each triangle is given. Write an equation and solve it to find x.

a)



The area is 24 cm². Find x.

b)



The area is 42 m². Find x.

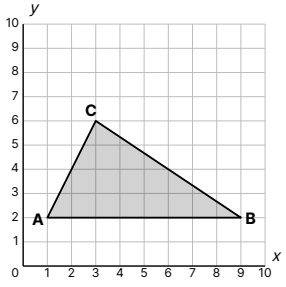
E Problems ●●●

6. A triangular sail has a base of 5 m and a height of 7 m. Sailcloth costs $28 per square metre. Find the cost of the cloth for the sail.

7. A rectangle 10 cm by 9 cm has a triangle cut from it. The triangle has its base along the whole bottom side of the rectangle and its top corner 4 cm up the left side. Find the area that is left.

8. A triangle has corners at A(1, 2), B(9, 2) and C(3, 6).

a) Find its area in square units.



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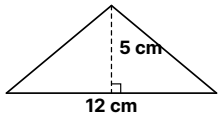
9. A triangular garden bed has a base of 16 m and a height of 9 m. One bag of mulch covers 4 m². How many bags are needed?

F Mixed practice ●●○

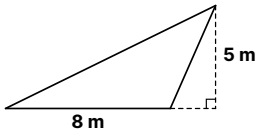
> **Remember:** Every kind of triangle from this sheet, in one go. Pick out the base and the perpendicular height first.

10. Find the area of each triangle.

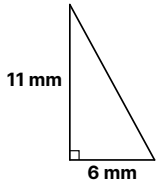
a)



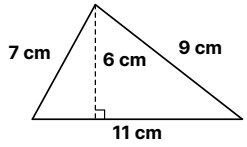
b)



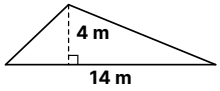
c)



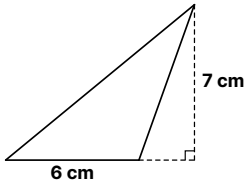
d)



e)

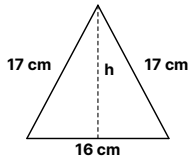


f)



Extension

11. An isosceles triangle has two sides of 17 cm and a base of 16 cm. Find its height first, then its area.



12. An equilateral triangle has sides of 12 cm. Find its area, to 1 decimal place.

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